

Is there more to aging than mitochondrial DNA and reactive oxygen species?

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Summary With the aging of the population, we are seeing a global increase in age-related disorders, especially in developed countries. Chronic diseases disproportionately affect the older segment of the population, contributing to disability, a diminished quality of life, and an increase in healthcare costs. Increased life expectancy reflects the success of contemporary medicine, which must now respond to the challenges created by this achievement, including the growing burden of chronic illnesses, injuries, and disabilities. A well-developed theoretical framework is required to understand the molecular basis of aging. This, in turn, is a prerequisite for developing the clinical interventions that will constitute an efficient response to the challenge of age-related health issues. This review critically analyzes the experimental evidence that supports and refutes the Free Radical/Mitochondrial Theory of Aging, which has dominated the field of aging research for almost half a century.